

MATH 3680 — L^AT_EX — Week 1

Arjan Singh Ellingson

August 24th, 2026

Below is the series expansion for e^x .

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}$$

This is the first 10 digits of π which is approximately $\frac{22}{7}$.

$$\pi \approx 3.1415926535$$

This is the quadratic formula which gives the solutions to the equation $ax^2 + bx + c = 0$.

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Slope intercept form is also useful when trying to plot lines only given the slope, m , and a point (x, y)

$$y = mx + b$$

And, lastly, I wanted to include the famous Pythagorean Theorem, which is

$$a^2 + b^2 = c^2$$

and allows you to compute $c = \sqrt{a^2 + b^2}$, where c is the hypotenuse from legs a and b .

Thanks for reading!!